MIL-I-14859(MU)
10 November 1970
SUPERSEDING
FA-PD-MI-2495
7 April 1961

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M14, ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M14, Assembly, loaded with Cartridge, Initiator, Delay, M84, for use in conjunction with other cartridge actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specifications for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms & Definitions
MIL-STD-414	- Sampling Procedures and Table for Inspection by Variables for Percent Defective
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

F10520861	- Initiator, Cartridge Actuated, Delay, M14, Assembly
C8846785	- Container, Ammunition, Fiber M459, For Initiators, Delay, Cartridge Actuated
C8846786	- Box Packing Ammunition, For Initiators, Delay, Cartridge Actuated, In Fiber Container, M459
IEL 10520861	- Inspection Equipment List For Initiator, Cartridge Actuated, Delay, M14, Assembly

(Copies of specifications, standards and drawings required for contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inch, minimum, and .425 inch, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator, shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, Size No. 4, as specified in MS28741.

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3.6.1 (Cont'd)

TABLE I

Temperature	Pressure, PSI	Ignition Delay (See 6.3)
-65° to -70°F	1300 to 2200	4.2 Seconds, Max.
65° to 75°F	1500 to 2200	3.0 ± .3 Seconds
160° to 165°F	1500 to 2200	2.3 Seconds, Min.

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship - Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr - The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.7.3 Foreign matter - The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions

4.2.1 Lot

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4.2.1.1 Submittal of product.- The contractor shall submit the product in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with acceptable cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	MAJOR Accept	MAJOR Reject	Minor Accept	Minor Reject
1- 300	100% Examination	-	-	-	-
301- 800	50	0	1	1	2
801- 1800	90	1	2	2	3

4.2.2.1 Cartridge, Initiator, Delay, M84, drawing D10520085, covering a detail of drawing F10520861.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing or inadequate	Visual

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4.2.2.2 Assembly, Dwg. F10520861, less all Dwgs except C10520410, C10520411 and MS 19060-3.

Defect	Method of Inspection
CRITICAL	
1. Firing pin protrusion	Gage
2. Steel ball missing	Gage
3. Striker hole in firing pin housing obstructed	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Dwg. F10520861, less Dwg. D10520085, D8597928, B10520496, C10520414 and A8359536.

CRITICAL	
1. Firing pin not cocked	Manual/Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Safety pin loose or deformed	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg. F10520861, less all Dwgs except D8597928, C10520414 and D10520085.

CRITICAL	
1. Cartridge not properly seated	Visual
2. Filter missing or insecure	Visual
3. Nylon patch missing or improperly applied	Visual

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Dwg. F10520861.

CRITICAL	
1. Cartridge face not seated against base of firing pin housing	X-ray 1/
2. Any part missing	X-ray 1/
3. Improper assembly	X-ray 1/
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Set screw missing or insecure	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Thread sealing compound missing from set screws	Visual
202. Shipping plug missing	Visual
203. Protective finish inadequate or defective	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.2.2.6 Container, Ammunition, Fiber M459 for Initiator, Delay, Cartridge Actuated, Dwg. C8846785.

CRITICAL

None defined

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4.2.2.6 (Cont'd)

Defect		Method of Inspection
MAJOR		
	None defined	
MINOR		
201.	Gap between body and cover, max.	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual
205.	Marking missing, incorrect or illegible	Visual
206.	Tape incomplete or badly wrinkled	Visual
207.	Length of tear tab, min.	Visual
208.	Color of tape incorrect	Visual

4.2.2.7 Box, Packing, Ammunition for Initiator, Delay, Cartridge Actuated in Fiber Container M459, Dwg. C8846786.

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Manual & Visual
202.	Hardware missing, loose, broken or improperly assembled	Manual & Visual
203.	Strapping missing, loose, broken or improperly assembled	Manual & Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of units incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

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4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each sub-assembly, consisting of all parts except the safety pin, cartridge and chamber, shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly consisting of all parts except the chamber and cartridge, shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirement of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirement of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional.- The following tests shall be performed by the Government:

4.2.3.5.1 Ballistics.- Twenty one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. The variables sampling plan of MIL-STD-414, example C-2 or C-3, as applicable, for the AQL listed in Table III shall be used for each sample of 7 units. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3, or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

Table III

Temperature	Sample Size	AQL
-65° to -70°F	7	0.65%
65° to 75°F	7	0.65%
160° to 165°F	7	0.65%

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4.2.4 Inspection equipment.- The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirements specified in 3.4 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned at the original temperature.

4.3.5.1.1 Temperature conditioning.- A standard M14 Initiator, loaded with a dummy M84 cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Packing and marking shall be in accordance with Drawings C8846785 and C8846786.

6. NOTES

6.1 Ordering data - Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is not a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

6.4 Supersession data - This specification includes the requirement of Purchase Description FA-PD-MI-2495 dated 7 April 1961.

Custodian:

Army - MU

Preparing activity:

Army - MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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